

Subject Description Form

Subject Code	ME39004/IC390
Subject Title	Applied Engineering Essentials: Python, Safety Practices, and Advanced CAD modelling
Credit Value	2 Training Credits
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	This subject offers a wide spectrum of fundamental engineering practice that are essential for a professional engineer. This subject includes Advanced CAD modelling CAD, Safety, Mechanism Design Practice and Scientific Computing Languages and it applications that aims at providing the technical skills to the students.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a) Interpret basic occupational health and industrial safety requirements for engineering practice; b) Apply scientific computing software for computing in science and engineering including visualization and programming; c) Apply advanced modeling techniques in CAD to effectively communicate technical information and documentation in accordance with engineering standards.

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Subject Synopsis/ Indicative Syllabus	Students are required to take 3 modules: <u>1) Industrial Safety</u> Safety Management: Overview, essential elements of safety management, safety training, accident management, and emergency procedures. Safety Law: F&IU Ordinance and principal regulations, OSH Ordinance and principal regulations. Occupational Hygiene and Environmental Safety: Noise hazard and control; dust hazard and control; ergonomics of manual handling. Safety Technology: Mechanical lifting, fire prevention, dangerous substances and chemical safety, machinery hazards and guarding, electrical safety, first aid, job safety analysis, fault tree analysis, and personal protective equipment. <u>2) AI application with Python</u> Problems solving with Python: Functions and Python packages to solve engineering problem (i.e. plot displacement diagram). AI application: Develop applications that leverage AI techniques for data manipulation and visualization, including the use of libraries for machine learning, data processing, and graphical user interfaces. <u>3) Computer Aided Design</u> Assembly Modeling: Learn to create and manage assemblies using both bottom-up approaches, explore various mate types to define component relationships, understand the management of subassemblies for complex projects. Advanced Part Modeling: Utilize advanced features such as lofts, sweeps, and patterns for complex geometry creation, implement design tables for efficient parametric modeling, and apply optimization strategies to enhance part performance and manufacturability. Surface Modeling: Gain an understanding of surface modeling fundamentals and applications in product design and explore methods to convert surface models into solid bodies for integration into assemblies.
Learning Methodology	The learning and teaching methods include lectures, workshop tutorials, and practical works. The lectures are aimed at providing students with an overall and concrete background knowledge required for understanding key issues in engineering communication, use of standard engineering components and systems, and importance of industrial safety. The workshop tutorials are aimed at enhancing students' in-depth knowledge and ability in applying the knowledge and skills to complete specific tasks. The practical works aim at facilitating students to review the diverse topics covered in this course and perform active learning with research, practice, questioning, and problem solving in a unified activity.

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Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <tr> <th rowspan="2">Assessment Methods</th> <th rowspan="2">Weighting (%)</th> <th colspan="3">Intended Learning Outcomes Assessed</th> </tr> <tr> <th>a</th> <th>b</th> <th>c</th> </tr> <tr> <td>1. Assignment</td> <td>80</td> <td>✓</td> <td>✓</td> <td>✓</td> </tr> <tr> <td>2. Test</td> <td>20</td> <td></td> <td>✓</td> <td></td> </tr> <tr> <td>Total</td> <td>100</td> <td></td> <td></td> <td></td> </tr> </table>				Assessment Methods	Weighting (%)	Intended Learning Outcomes Assessed			a	b	c	1. Assignment	80	✓	✓	✓	2. Test	20		✓		Total	100			
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			a	b	c																						
	1. Assignment	80	✓	✓	✓																						
	2. Test	20		✓																							
Total	100																										
Assignments in the form of system configuration or fabrication tasks will be used to assess how well students understand the working principle, capabilities, and operation of the aviation systems and processes. Students' skill-level will be evaluated by the artifacts they produced, while their engineering judgment and critical thinking be evaluated by individually filled task worksheets. Test is designed to facilitate students to review the breadth and depth of their understanding on specific topics.																											
Student Study Effort Expected	Class Contact																										
	Min-lecture /Hands-on Practice/ Product Assembly /Report Writing			60 Hrs																							
	Other Study Effort			0 Hrs.																							
	Total Study Effort			60 Hrs.																							
Reading List and References	Reference Software List: 1. SolidWorks from Dassault Systèmes Solidworks Corp. 2. Python from Python Software Foundation Reference Standards and Handbooks: 1. BS EN ISO 128 – Technical product documentation. General principles of representation Reference Books: 1. Joshi, P. (2017). <i>Artificial intelligence with python</i>. Packt Publishing Ltd. 2. Training material, manual and articles published by Industrial Centre.																										